Task 1: Ask for Location Permission

```
import Foundation

enum LocationStatus {
    case notDetermined
    case denied
    case restricted
    case authorizedWhenInUse
}

func showLocationStatus(_ status: LocationStatus) {
    switch status {
    case .notDetermined:
        print("Location permission has not been requested.")
        print("Requesting location permission...")

    case .denied:
        print("Location permission denied.")
        print("Enable location access in Settings.")

    case .restricted:
        print("Location access is restricted.")

    case .authorizedWhenInUse:
        print("Location access granted.")
```

```
        print("Location updates can begin.")
    }
}

showLocationStatus(.notDetermined)
print()
showLocationStatus(.denied)
print()
showLocationStatus(.restricted)
print()
showLocationStatus(.authorizedWhenInUse)
```

Output

```
Location permission has not been requested.
Requesting location permission...

Location permission denied.
Enable location access in Settings.

Location access is restricted.

Location access granted.
Location updates can begin.
```

Task 2: Camera Permission and Photo Selection

```
import Foundation
func openCamera(hasUsageDescription: Bool) {
    if hasUsageDescription {
        print("Camera usage description found.")
        print("Camera can be opened on a supported device.")
    } else {
        print("Camera usage description is missing.")
        print("The app may terminate when camera access is requested.")
    }
}

func selectPhoto() {
    print("Photo picker opened.")
    print("User can select a photo without requesting photo library permission.")
}

print("Test 1: Camera permission")
openCamera(hasUsageDescription: false)

print("\nTest 2: Camera permission added")
openCamera(hasUsageDescription: true)

print("\nTest 3: Photo selection")
selectPhoto()
```

Output

```
Test 1: Camera permission
Camera usage description is missing.
The app may terminate when camera access is requested.

Test 2: Camera permission added
Camera usage description found.
Camera can be opened on a supported device.

Test 3: Photo selection
Photo picker opened.
User can select a photo without requesting photo library permission.
```

Task 3: Minimum OS Version and Target Devices

```
import Foundation

let deploymentTarget = 16

let requiredFeatureVersion = 17

print("Deployment target: iOS \(deploymentTarget)")

print("Required feature: iOS \(requiredFeatureVersion)")


if deploymentTarget < requiredFeatureVersion {

    print("Error: The feature requires a newer iOS version.")

    print("Fix: Increase the deployment target to iOS 17 or use availability checks.")

} else {

    print("Deployment target supports the feature.")

}
```

```
let supportsIPhone = true
let supportsIPad = true

print("\niPhone support: \(supportsIPhone ? "Enabled" : "Disabled")")
print("iPad support: \(supportsIPad ? "Enabled" : "Disabled")")
```

Output

```
ERROR!
Deployment target: iOS 16
Required feature: iOS 17
Error: The feature requires a newer iOS version.
Fix: Increase the deployment target to iOS 17 or use availability
checks.

iPhone support: Enabled
iPad support: Enabled
```